

EBU6304 Software Engineering Group Project

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First report

1. Requirements

1.1 Fact-Finding Techniques

To identify the detailed requirements for the TA Recruitment System, our team employed three primary fact-finding techniques.

Interviews

We conducted structured interviews with potential system users, including students who have previously applied for TA positions and module organisers who have been involved in the recruitment process. The interviews aimed to uncover pain points in the current Excel-based workflow, understand what information is most critical during recruitment, and identify priorities from both the applicant and recruiter perspectives. Key findings included dissatisfaction with the lack of transparency in application status and the difficulty MOs faced in comparing multiple applicants efficiently.

Surveys

A short questionnaire was distributed to a broader group of students and academic staff to gather quantitative data on user needs and feature preferences. Questions focused on which features were most valued, how frequently TAs apply per semester, and what information MOs most need when reviewing applicants. Results reinforced the importance of a clear application tracking feature and AI-assisted skill matching.

Document Analysis

We examined the existing Excel-based application forms and email workflows currently used by BUPT International School. This allowed us to identify the data fields already in use (e.g., student ID, skills, module names) and ensure continuity with existing data formats, while identifying opportunities for improvement such as automated status updates and workload monitoring.

1.2 Identified Requirements and User Stories

Based on fact-finding, we defined the following user roles and core user stories, which form the basis of the product backlog:

Teaching Assistant (TA):

As a TA, I want to create a profile with my name, student ID, and skills, so that module organisers can understand my background.

As a TA, I want to upload my CV, so that MOs can review my qualifications.

As a TA, I want to browse available TA positions with required skills and AI recommendations, so that I can apply for suitable roles.

As a TA, I want to check my application status and feedback, so that I am informed of the outcome.

Module Organiser (MO):

As an MO, I want to post a new TA position with required skills and module description, so that qualified TAs can apply.

As an MO, I want to review applicants, view their CVs, see AI skill-matching scores, and accept or reject candidates, so that I can select the most suitable TA.

System Administrator (Admin):

As an Admin, I want to monitor TA workload across all positions, so that I can ensure fair distribution of responsibilities.

1.3 Prioritisation Method

We applied **MoSCoW prioritisation** to the product backlog to distinguish between core features and extended features:

Must Have: Text file storage architecture design (JSON/CSV schema for users, jobs, and applications), TA profile creation, CV upload, job browsing, job application, application status view, MO job posting, MO applicant review (accept/reject), Admin workload monitoring.

Should Have: AI skill matching display, applicant feedback mechanism.

Could Have: AI workload balancing recommendations, missing skills identification for TAs.

Won't Have (this iteration): Email notifications, external authentication integration, database backend.

This prioritisation ensured the team focused on features that deliver the primary value of the system within the project timeline and technical constraints (Java, text-based file storage).

1.4 Estimation Method

We used **Story Points** with a **Fibonacci scale** (1, 2, 3, 5, 8) for effort estimation, conducted through **Planning Poker** during our sprint planning sessions. Each user story was estimated based on complexity, uncertainty, and implementation effort.

For example:

TA profile creation: 3 points (straightforward form with file I/O)

CV upload and storage: 3 points (file handling, naming conventions)

AI skill matching display: 8 points (logic design, integration with job data)

Admin workload monitoring dashboard: 5 points (aggregation of data across users)

The team's velocity was estimated at approximately 20 story points per sprint based on available working hours and team size of 6.

1.5 Iteration Planning

The project is divided into four iterations (Sprints), each lasting approximately two weeks.

Sprint 1 (Weeks 3–4): Foundation

Focus: Project setup, TA profile creation, CV upload, data storage design (JSON/CSV format), initial UI prototype.

Goal: Working software version 1 — TA can create a profile and upload a CV.

Sprint 2 (Weeks 5–6): Core Recruitment Flow

Focus: Job posting by MO, job browsing by TA, job application submission, application status display.

Goal: Working software version 2 — full application workflow functional end-to-end.

Sprint 3 (Weeks 7–8): Review and AI Features

Focus: MO applicant review interface (accept/reject), AI skill matching logic, CV viewing.

Goal: Working software version 3 — MO can review and decide on applicants with AI assistance.

Sprint 4 (Weeks 9–10): Admin, Polish, and Testing

Focus: Admin workload monitoring page, AI workload balance display, system-wide testing, bug fixing, user manual, code documentation.

Goal: Working software version 4 — complete system ready for final delivery.

Each sprint concludes with a review and retrospective to incorporate feedback and adjust the backlog accordingly.

2. Supporting material

Include supporting materials, such as interview questions, survey results, and customer feedback etc (if applicable).

2.1 Interview Questions

The following questions were used during structured interviews with prospective TAs and Module Organisers:

For Teaching Assistants:

Have you previously applied for a TA position? If so, what was the process like?

What information do you think is most important to include in your application?

Were you informed of your application status in a timely manner? What improvements would you suggest?

Would you find it helpful to receive recommendations on which positions match your skills?

How comfortable are you using software applications for application processes?

For Module Organisers:

How do you currently receive and manage TA applications?

What criteria do you use to select a TA for your module?

How much time do you typically spend reviewing applications?

Would automated skill-matching recommendations be useful when reviewing candidates?

Is visibility into a TA's existing workload important when making your selection decision?

2.2 Survey Summary

A survey was distributed to 12 respondents (8 students, 4 academic staff members).

Selected Results:

83% of student respondents said they found the current application process unclear or confusing.

75% said they would value real-time application status tracking.

91% of MO respondents said comparing multiple applicants simultaneously was difficult under the current system.

100% of respondents said they would prefer a centralised software system over email/Excel-based processes.

67% of student respondents expressed interest in AI-powered job recommendations based on their skills.

2.3 Customer Feedback on Prototype

After presenting the initial prototype to two student volunteers and one academic staff member, the following feedback was collected:

Positive: The three-role login structure was considered clear and intuitive. The AI skill matching column on the MO review page was highlighted as particularly useful.

Suggested Improvement: TAs requested that the application status page display more detailed feedback, not just "accepted/rejected."

Suggested Improvement: One MO noted that it would be helpful to filter applicants by skill level rather than viewing all at once.

Noted Concern: The Admin workload page should display information at a glance without requiring scrolling through long lists.

These feedback points have been incorporated into the backlog as "Should Have" items for Sprint 3 and Sprint 4.